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Passport: 23AA94614 **Work permit:** Georgian **Gender:** Female **Date of birth:** 08/02/2002 **Place of birth:** Tbilisi, Georgia **Nationality:** Georgian

ABOUT ME

Dedicated and highly motivated individual pursuing a career in Automation Control Systems (ECE), consistently striving to deepen proficiency in the field. Eager to broaden knowledge in main field, as well as in Computer Science, with a focus on mastering essential coding languages. Proficient in various design programs, with a track record of successfully completed multiple project during academic tenure. I leverage problem-solving skills and a determined mindset to deliver successful outcomes. Possessing exceptional communication skills and interpersonal skills, I excel both independently and as part of a team. Capable of assuming leadership roles when necessary. I'm committed to leveraging all acquired knowledge and skills to excel as an engineer and contribute effectively to the field.

WORK EXPERIENCE

FZJ (Forschungszentrum Jülich) - ICA (Integrated Computing Architectures)

City: Tbilisi | **Country:** Georgia

[01/02/2026 – Current]

Engineering researcher

Master's Thesis Research with GGSB26 Master's Program.

Thesis: "Scalable Research Data Management for High-Speed Instrumentation":

- Integrating database and software layers for high-speed detector readout system;
- Developing FPGA-based data acquisition firware on ZCU104 (Zynq Ultrascale+);
- Designing AXI4-Lite register interfaces and custom IP cores in VHDL/Vivado.

Caucasus University

City: Tbilisi | **Country:** Georgia

[01/09/2025 – Current]

Lecturer in electrical engineering

- Delivered core Analog and Digital Circuits topics in native language to undergraduate students.
- Delivered core Microprocessors and Microcontroller topics in native language to undergraduate students.
- Developed and managed course materials, assessments and exams.
- Graded student work and provided constructive feedback.

University of Georgia

City: Tbilisi | **Country:** Georgia

[20/03/2025 – Current]

Lecturer in electrical engineering

- Delivered core Circuit Analysis topics in English to undergraduate students.
- Delivered core Analog Circuits topics in English to undergraduate students.
- Delivered core Microprocessors and Microcontroller topics in English to undergraduate students.
- Developed and managed course materials, assessments and exams.
- Graded student work and provided constructive feedback.

Forschungszentrum Jülich (FZJ) - Integrated Computing Architectures (ICA)

City: Tbilisi | **Country:** Georgia

[22/11/2024 – 31/12/2025]

FPGA Engineer

- Executed FPGA-based projects using Vivado design suite.
- Utilized FGPA platforms and simulation tools for development and testing.

Helmholtz-Zentrum Dresden-Rossendorf (HZDR)

City: Dresden | **Country:** Germany

[02/04/2024 – 15/05/2024]

Intern

- Developed and tested software for 32-bit microcontrollers using STMCubeIDE.
- Worked on stepper motor control, focusing on programming and testing with stepper motor drivers and Nucleo STM Boards.
- Gained hands-on experience in embedded systems programming and hardware integration.

Independent

City: Tbilisi | **Country:** Georgia

[2023 – 2024]

Private tutor

- Delivered personalized English lessons tailored to individual student needs.
- Incorporated diverse activities and multiple resources to enhance students' language skills.
- Focused on development of vocabulary and grammar, while refining reading, writing, listening, and speaking skills.
- Monitored student progress and adapted teaching methods to maximize learning outcomes.

Agricultural University of Georgia

City: Tbilisi | **Country:** Georgia

[14/10/2021 – 14/02/2022]

TA

- Consulting on various activities, such as debates and test papers, which involves preparing students by highlighting key details pertinent to each specific activity;
- Reviewing the outcomes of these activities and assigning ratings based on a predefined scoring scale aligned with the syllabus.

EDUCATION AND TRAINING

[30/09/2024 – Current]

Master's in Automation Control Systems (ECE Masters)

Agricultural University of Georgia https://agruni.edu.ge/en/

City: Tbilisi | **Country:** Georgia | | **Level in EQF:** EQF level 7

[31/08/2020 – 17/09/2024] **Bachelor's Degree in Automation Control Systems (ECE Bachelors)**

Agricultural University of Georgia https://agruni.edu.ge/en/

City: Tbilisi | **Country:** Georgia | **Field(s) of study:** Engineering, manufacturing and construction: ● *Electronics and automation* ● *Electricity and energy* | **Final grade:** 3.52/4.0 | **Level in EQF:** EQF level 6 | **NQF Level:** 6 | **Type of credits:** ECTS | **Number of credits:** 240 | **Thesis:** Project-based coursework; No thesis required.

- Core courses in circuit analysis, analog and digital electronics, microcontroller-based (AVR, STM32) system design and control systems.
- Developed practical skills in embedded systems, FGPA-based design, and hardware simulation.
- Gained experience with programming languages (C/C++, Python, Assembly, MATLAB, VHDL).
- Used simulation environment tools (NI Multisim, Proteus, Logisim, MATLAB).
- Worked in various coding environments (Visual Studio, Microchip Studio, AVR Studio, FlowCode, STM32CubeIDE, STM32CubeMX, STM32Keil, Xilinx).
- Trained in CAD and 3D modeling using AutoCAD and Blender.
- Acquired basis knowledge in computer networking principles and protocols (TCP/IP, UDP, UTP, DNS, DHCP)

[2008 – 2020] **Secondary Education**

Private School LTD "'Shovi"

City: Tbilisi | **Country:** Georgia | | **Level in EQF:** EQF level 4

Graduated with golden medal.

LANGUAGE SKILLS

Mother tongue(s): Georgian

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** C1

SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** C1

Russian

LISTENING B2 **READING** B1 **WRITING** B1

SPOKEN PRODUCTION B1 **SPOKEN INTERACTION** B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

CONFERENCES AND SEMINARS

[26/06/2023 – 30/06/2023] **GGSB Lecture Week** Tbilisi

- The lecture week was held at the Universities of the Consortium (AUG, GTU, TSU) on June 26-30, 2023 and was a joint activity of Forschungszentrum Jülich (FZJ), GSI Darmstadt, Helmholtz-Zentrum Dresden-Rossendorf (HZDR) and the Georgian Universities (AUG, GTU, KIU, ISU and TSU) within the scope of the Georgian-German Science Bridge.
- Acquired knowledge across various engineering disciplines through interdisciplinary coursework.
- Delivered presentations on projects developed independently and in collaboration with AUG University.

[24/04/2023 – 28/04/2023] **Lecture course in "Measurements and Sensor Systems"** Tbilisi

- The lecture course in "Measurement and Sensor Systems" delivered by Prof. Dr. Stefan van Waasen, held at Kakha Bendukidze Campus on 24-28 April 2023.
- The course focused on sensor systems for measurement applications and the use of operational amplifiers in circuit design.

HONOURS AND AWARDS

[30/06/2023] **Certificate Awarding institution:** Georgian-German Science Bridge (GGSB)

- Acquired knowledge across various engineering disciplines through interdisciplinary coursework.
- Delivered presentations on projects developed independently and in collaboration with AUG University.

[28/04/2023] **Certificate Awarding institution:** Forschungszentrum Jülich (FZJ) ZEA-2

- The lecture course in "Measurement and Sensor Systems" delivered by Prof. Dr. Stefan van Waasen, held at Kakha Bendukidze Campus on 24-28 April 2023.
- The course focused on sensor systems for measurement applications and the use of operational amplifiers in circuit design.

PROJECTS

[01/09/2025 – 31/01/2026] **Talking Clock for Visually Impaired and Blind**

- Designed and implemented a Georgian talking clock for visually impaired and blind users using an 8-bit ATmega324P microcontroller with DS1307 RTC module, MP3 Player Mini (UART) and 16x2 LCD display;
- Validated the circuit in Proteus Professional and developed the control algorithm in Flowcode and CVAVR prior to physical implementation.
- The project was subsequently migrated to an STM32 microcontroller, significantly expanding its capabilities.

[01/03/2024 – 31/07/2024] **Robotic Manipulator Arm**

- Designed and implemented a 6-DOF robotic manipulator arm controlled by an 8-bit ATmega microcontroller, with each joint driver by MG996R high-torque servo motors;
- Implemented 6-channel PWM signal generation (50 Hz, 1-2.5 ms pulse width) to control servo joint angles proportionally to potentiometer input read via ADC;
- Validated the circuit in Proteus Professional and developed the control algorithm in Flowcode and Microchip Studio 7 prior to physical implementation.

[01/09/2023 – 28/12/2023] **Three-Phase Voltage and Phase Shift Meter with Automatic Phase Switcher**

- Designed and implemented a three-phase voltage monitoring and automatic phase-sequencing system center on an ATmega168 microcontroller, measuring per-phase voltages via ADC and phase shift angles via zero-crossing pulse detection with a 16-bit timer/counter;
- Implemented automatic phase-sequence correction dual-relay logic and a built-in protection system that disconnects the load upon voltage deviation exceeding 7% or out-of-threshold conditions;

- Designed pulse-shaping circuit using PC817 optocouplers, 76HC14 Schmitt trigger, and RC differentiators;
- Validated the circuit in Proteus Professional and developed the control algorithm in Flowcode and Microchip Studio 7 prior to physical implementation.

[01/03/2023 – 30/06/2023] **Obstacle-Avoiding Car**

- Designed and implemented an autonomous obstacle-avoiding vehicle.
- The car uses an HC-SR04 ultrasonic sensor to detect obstacles in its path, an AVR 8-bit microcontroller for decision-making logic, and a motor driver to control movement. When detecting an obstruction, the vehicle stops and reroutes in the direction of least resistance.
- Evaluated sensing technologies (ultrasonic, IR, LiDAR, radar) and path-planning algorithms (Potential Field Method, Bug Algorithm, Dynamic Window Approach) to justify design choices.
- Validated the circuit in Proteus Professional and developed the control algorithm in Flowcode and Microchip Studio 7 prior to physical implementation.

[02/09/2022 – 28/12/2022] **Toy Car Control with Two Motors and Collision Avoidance System**

- Designed and implemented a two-motor-driven toy car with an integrated collision avoidance system. T
- The project involved DC motor control via an H-bridge driver (L298N), infrared distance sensing for obstacle detection, and microchip-based logic to automatically halt or redirect the vehicle upon detecting objects within a defined proximity threshold.

[01/03/2021 – 31/07/2021] **Wall Cable Detector**

- The device detects the electromagnetic field emitted by live cables to identify their location within walls;
- Design and simulation in-wall electrical cable detection device capable of locating hidden wiring without physical wall damage;
- The project covered the operating principles of electrostatic and electromagnetic detection methods, comparative analysis of detector types (electrostatic, electromagnetic and combined), circuit schematic design, electronic component selection and calculations, and a full circuit simulation in Proteus Professional.